

Multilevel Models 2025/26

Model selection in multilevel models

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December 19, 2025

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Model setup: Linear mixed-effects model (LMM)

For groups $i = 1, \dots, m$:

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i + \boldsymbol{\varepsilon}_i$$

$$\mathbf{b}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{D}), \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_{n_i})$$

- $\mathbf{y}_i \in \mathbb{R}^{n_i}$ response vector in group i
- $\mathbf{X}_i$ fixed-effects design, $\boldsymbol{\beta}$ fixed parameters
- $\mathbf{Z}_i$ random-effects design, $\mathbf{b}_i$ random effects
- Assumptions (typical LMM):
- Marginal variance: $\boldsymbol{\Omega}_i = \mathbf{Z}_i\mathbf{D}\mathbf{Z}_i^\top + \sigma^2\mathbf{I}_{n_i}$

Model family

Model 6 (varying intercept + varying slope):

$$y_{ij} \sim \mathcal{N}(\alpha_i + \beta_i \text{basement}_{ij}, \sigma^2),$$
$$\begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \gamma_0 + \gamma_1 \text{uranium}_i \\ \delta_0 + \delta_1 \text{uranium}_i \end{pmatrix}, \begin{pmatrix} \sigma_\alpha^2 & \rho \sigma_\alpha \sigma_\beta \\ \rho \sigma_\alpha \sigma_\beta & \sigma_\beta^2 \end{pmatrix} \right).$$

alternatives

- Model 0: intercept only
- Model 1: fixed effect of basement only (no random effects)
- Model 2: random intercept only
- Model 3: random intercept+slope, no uranium terms
- Model 4: like Model 6 but **no cross-level interaction** ($\delta_1 = 0$)
- Model 5: like Model 6 but **uncorrelated** random effects ($\rho = 0$)

Problem 1

Researchers need tools for:

- **Model adequacy** (how well does the model explain / predict?)
 - **A**: overall model adequacy (fixed + random)
 - **F**: adequacy due to fixed effects (useful for fixed-effect selection)
- **Model selection** (which fixed/random effects to keep?)
 - **S**: model selection (information criteria, etc.)

Key distinction:

- **Explained variation** extensions (“ R^2 -like”)
- **Concordance** (agreement between observed and predicted)
- **Information criteria** (penalized fit: AIC/BIC/cAIC/DIC)
- Also: some measures are **absolute** vs. **relative** (need a null model).

Measures:

Overall adequacy (A / A&S)

- **Zheng (2000)**: proportional reduction in deviance D_{rand} ; penalized variant P_{rand} .
- **Vonesh et al. (1996)**: concordance correlation r_c (marginal vs conditional); adjusted $r_{c,a}$.
- **Xu (2003), Liu et al. (2008)**: multiple R^2 extensions (often require null model).
- **Gelman & Pardoe (2006)**: Bayesian “level”-wise explained variation and pooling factors.

Model selection (S)

- **mAIC / BIC**: marginal likelihood based.
- **cAIC**: conditional likelihood + effective degrees of freedom (ML or REML variants).
- **DIC**: Bayesian deviance + effective parameters.

Simulation: results

- Within-category measures are often **highly correlated** (similar rankings), but:
 - Choice of **null model** can change what is being measured.
- For overall adequacy + selection (A&S):
 - P_{rand} stood out: consistently identified the population model (benefits from penalizing large random effects).
 - Conditional concordance $r_{c,a}$ is also attractive: no null model, fewer distributional assumptions, broader applicability.
- For pure model selection (S):
 - Among information criteria, **cAIC with REML** performed best in not missing the population model in their simulation.

Practical workflow

Steps:

- ① Select candidate model(s) with **cAIC REML**
- ② Confirm fixed part with **marginal** P_{rand} / $r_{c,a}$
- ③ Report overall adequacy with **conditional** P_{rand} / $r_{c,a}$

Interpretation note:

- P_{rand} : interpretable like an R^2 -style proportional reduction (penalized quasi-likelihood).
- $r_{c,a}$: interpretable as **concordance** (agreement between observed and predicted).

Problem 2

Problem 3